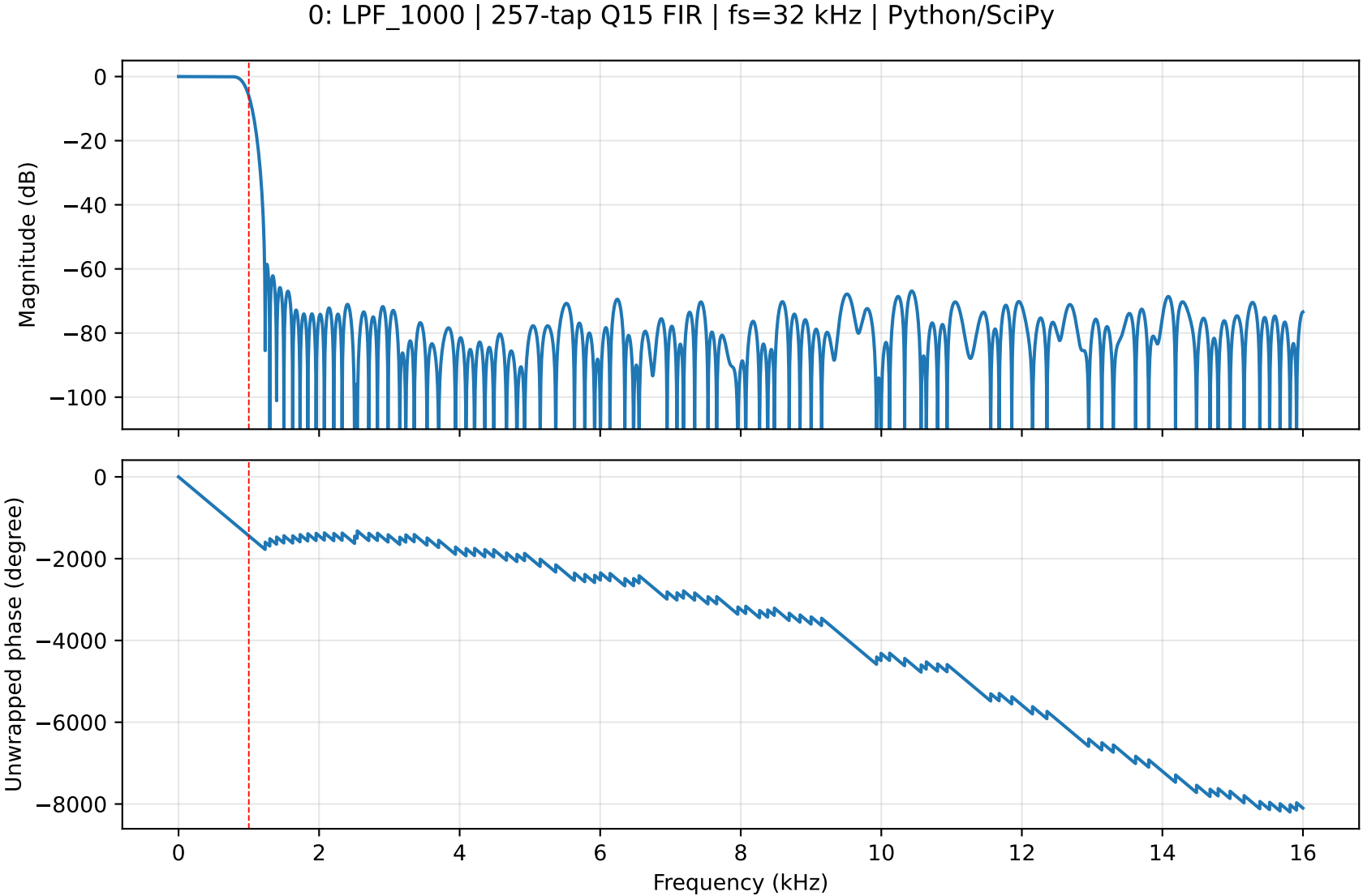
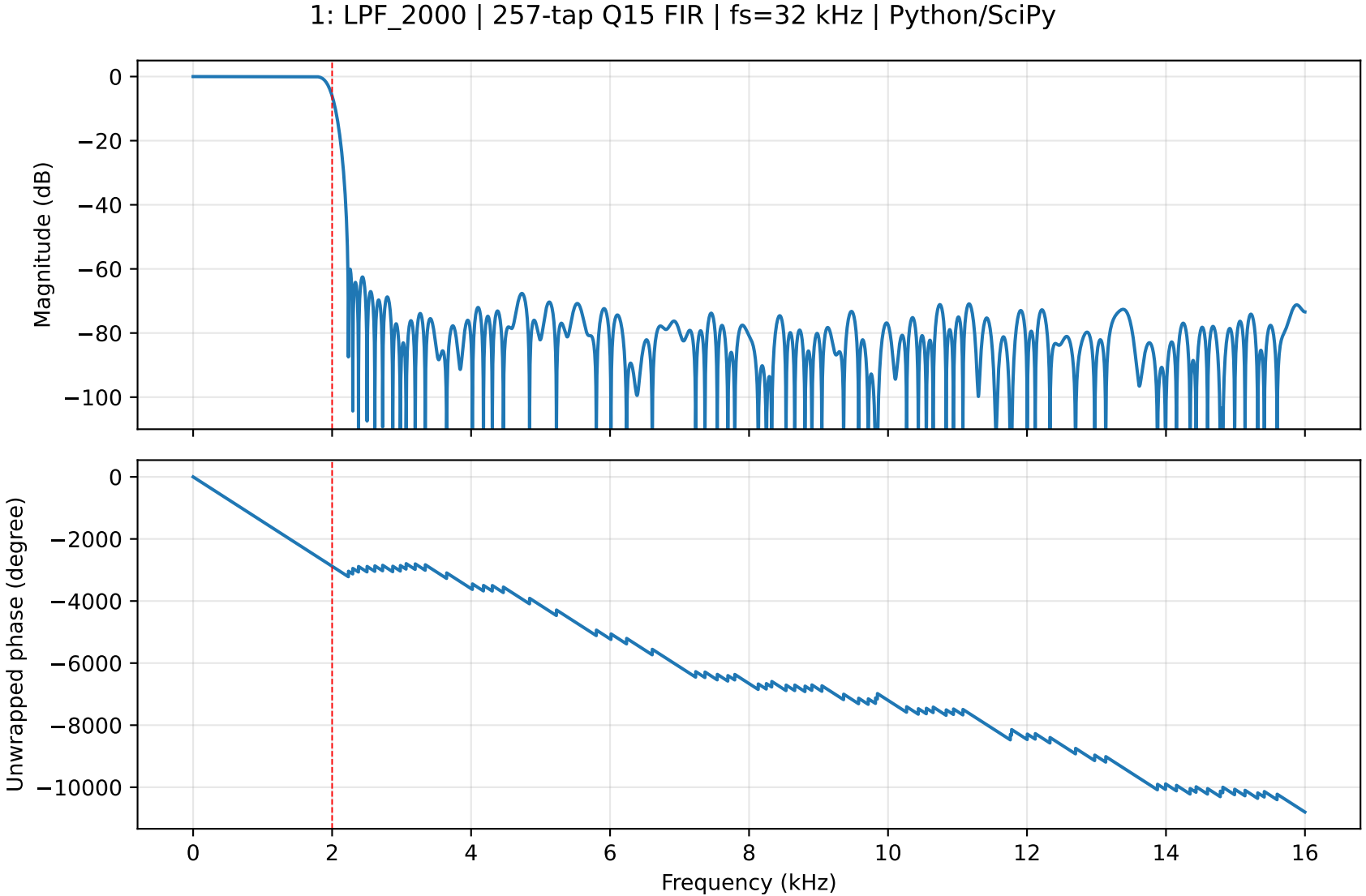
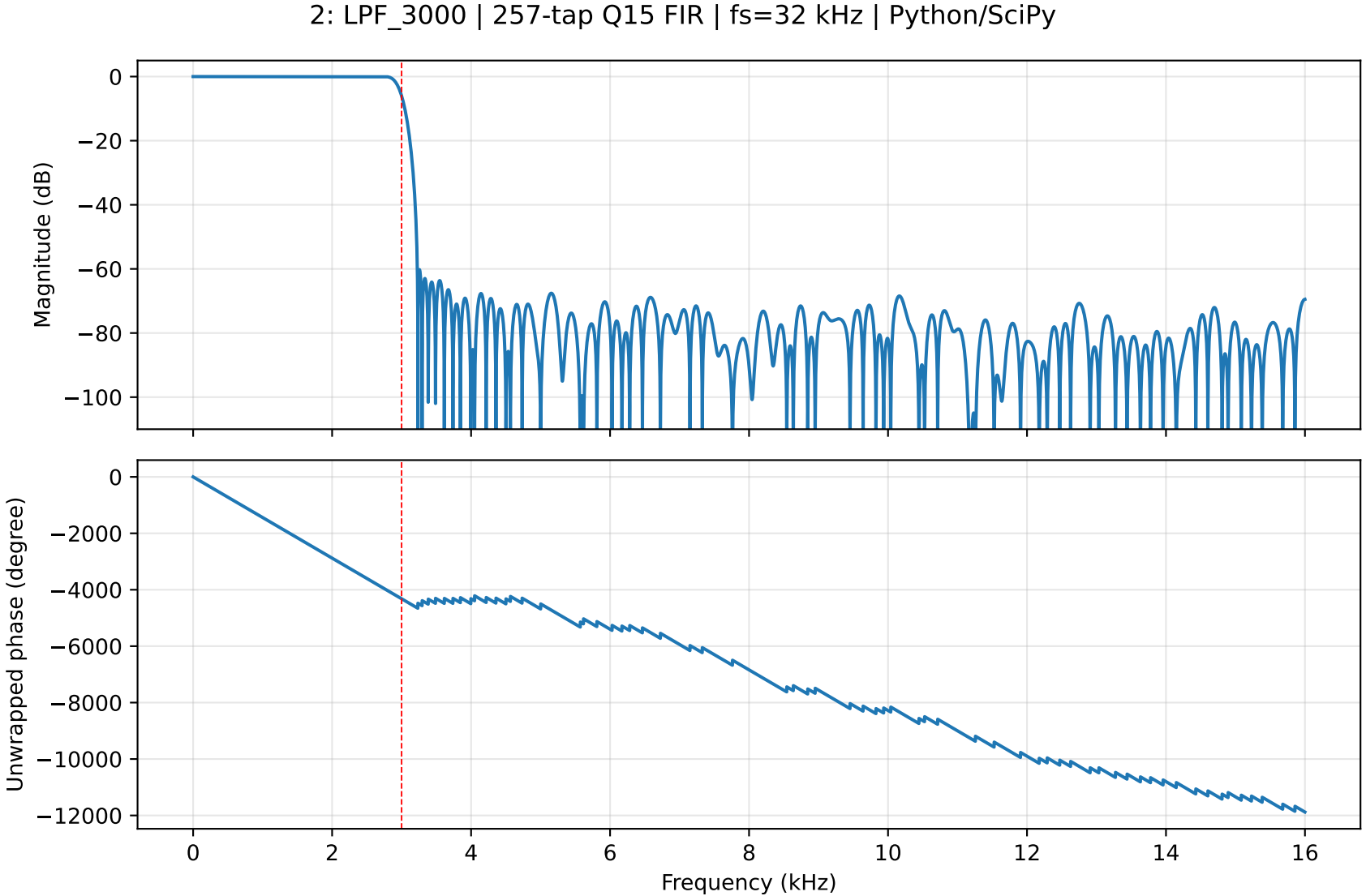
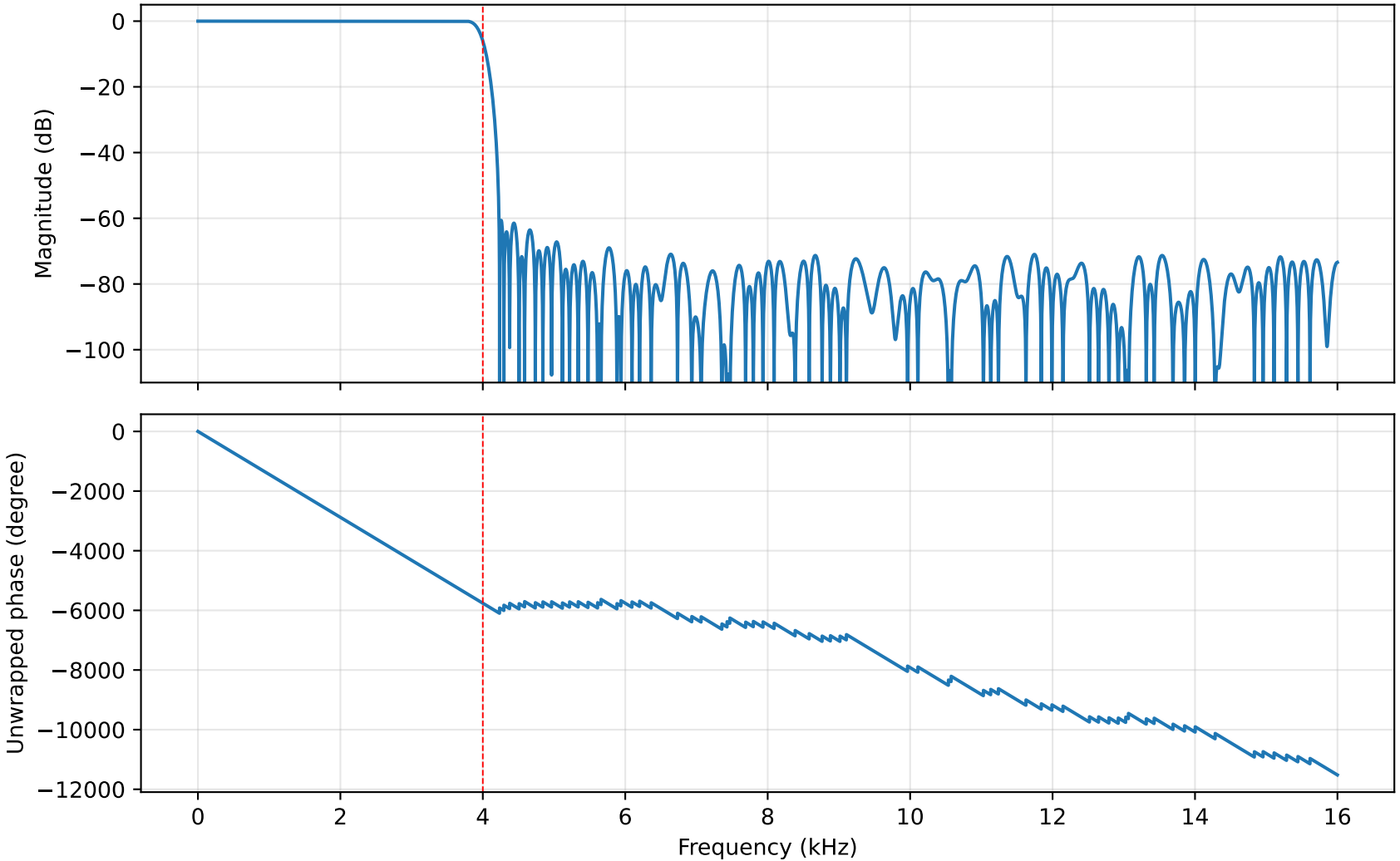
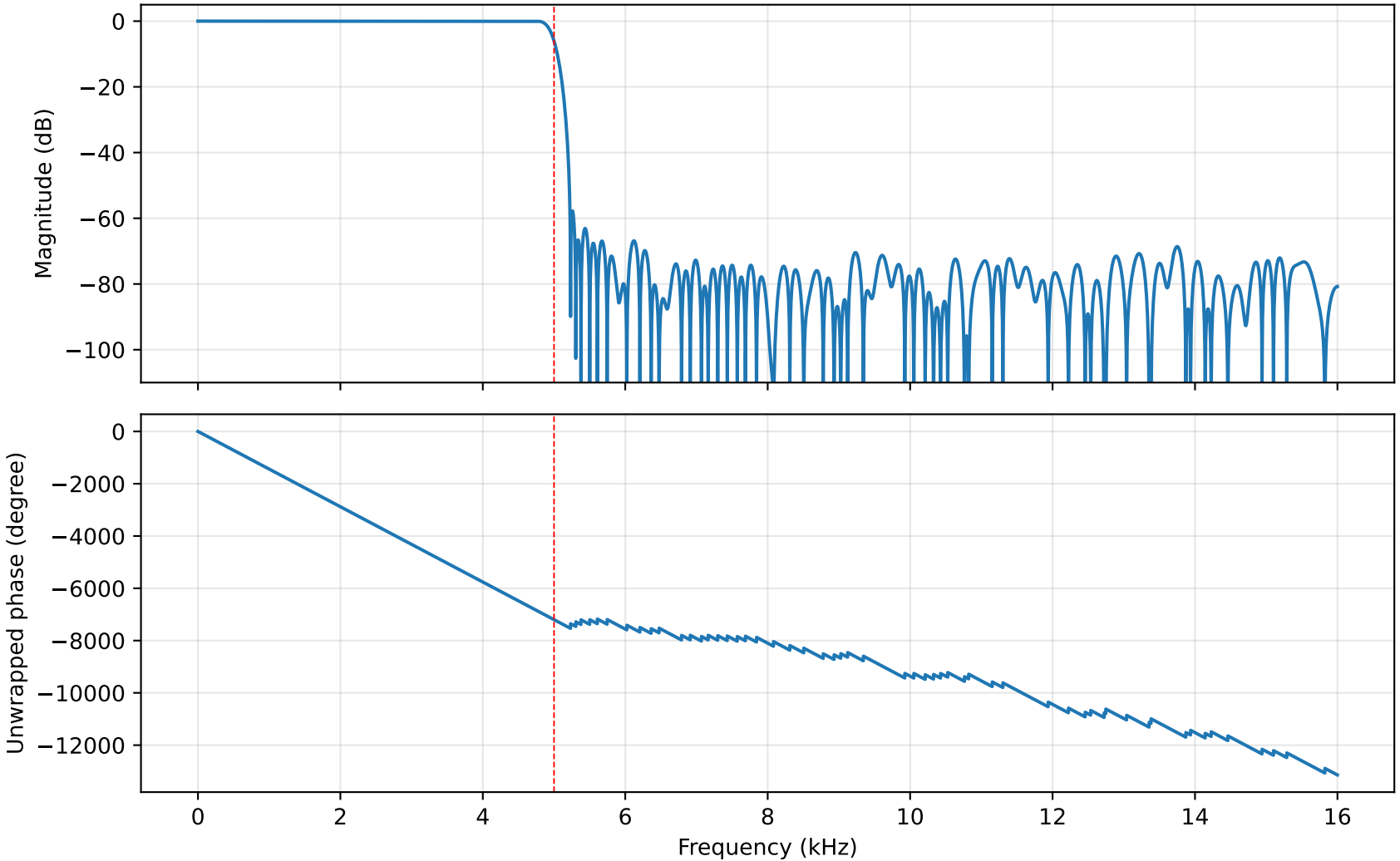
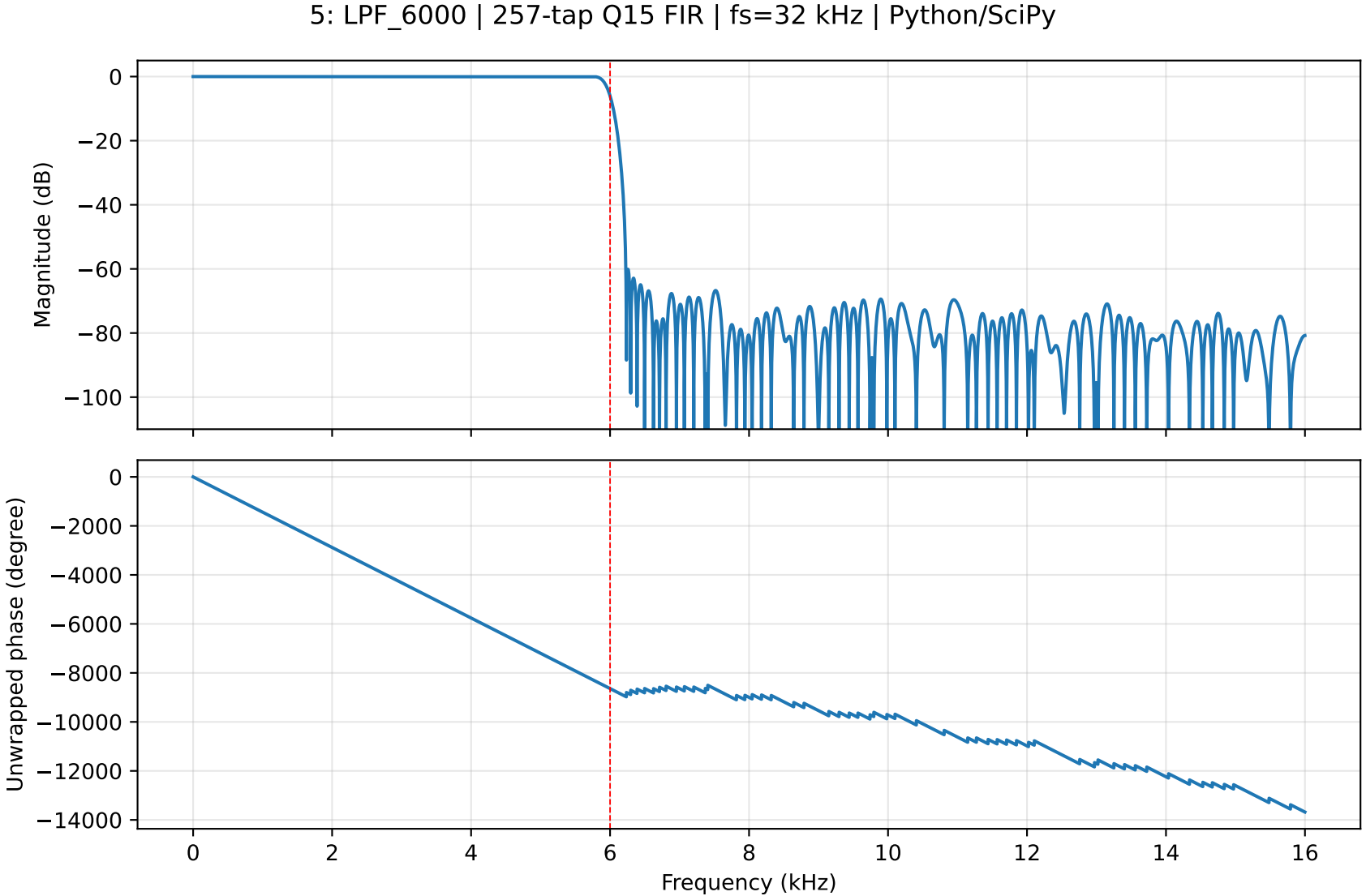

0: LPF_1000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

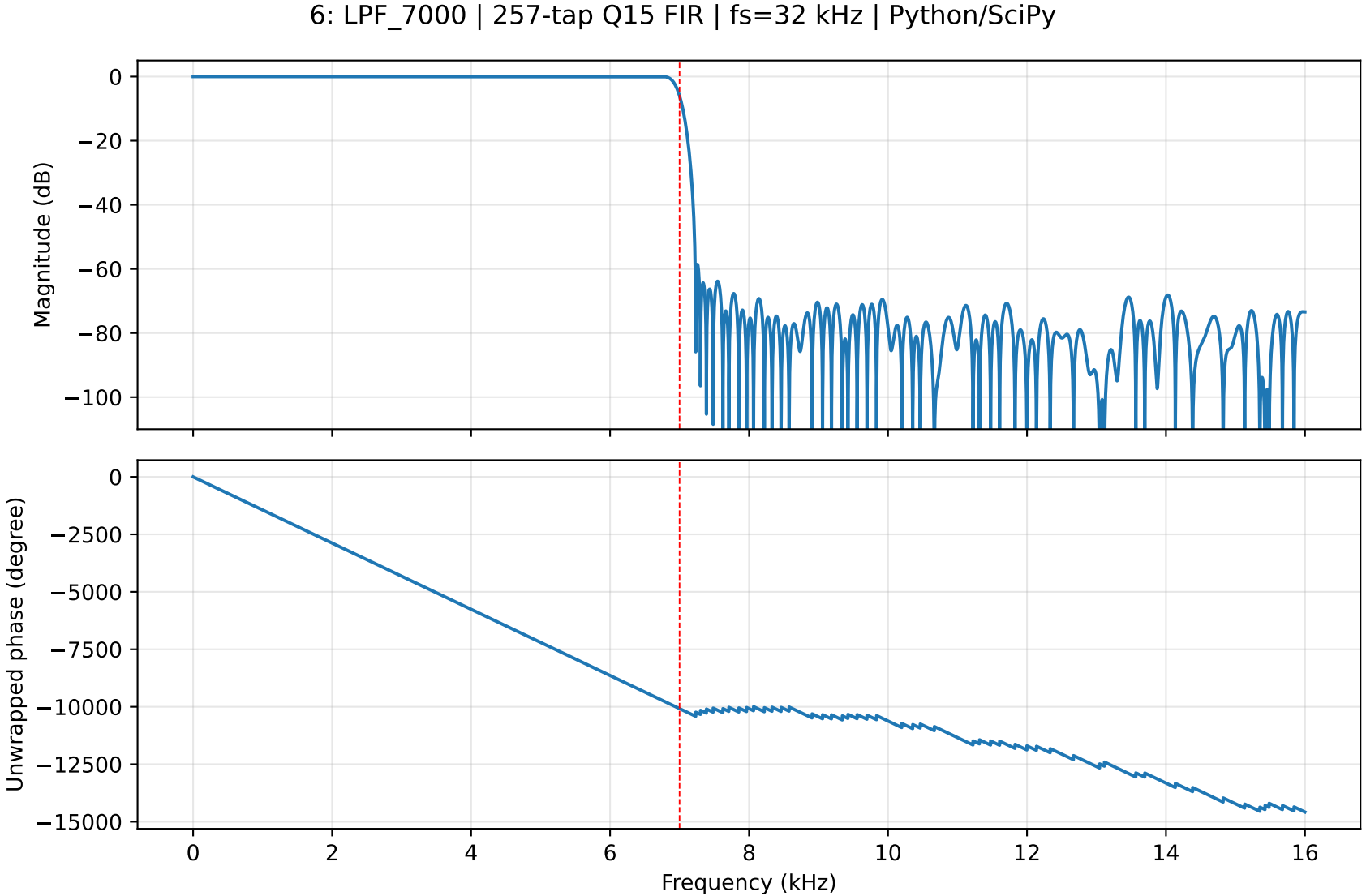

1: LPF_2000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

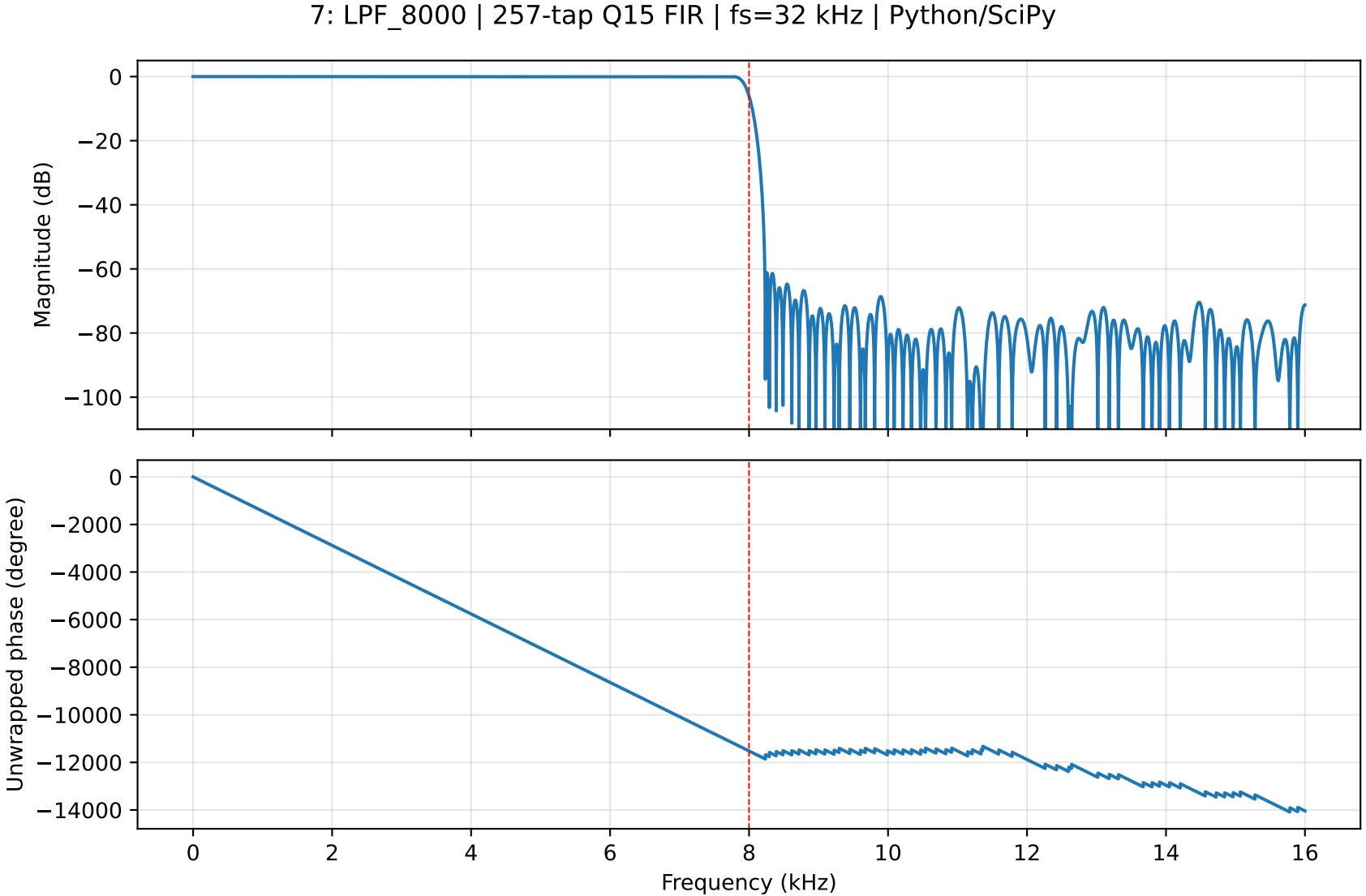

2: LPF_3000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
−12000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

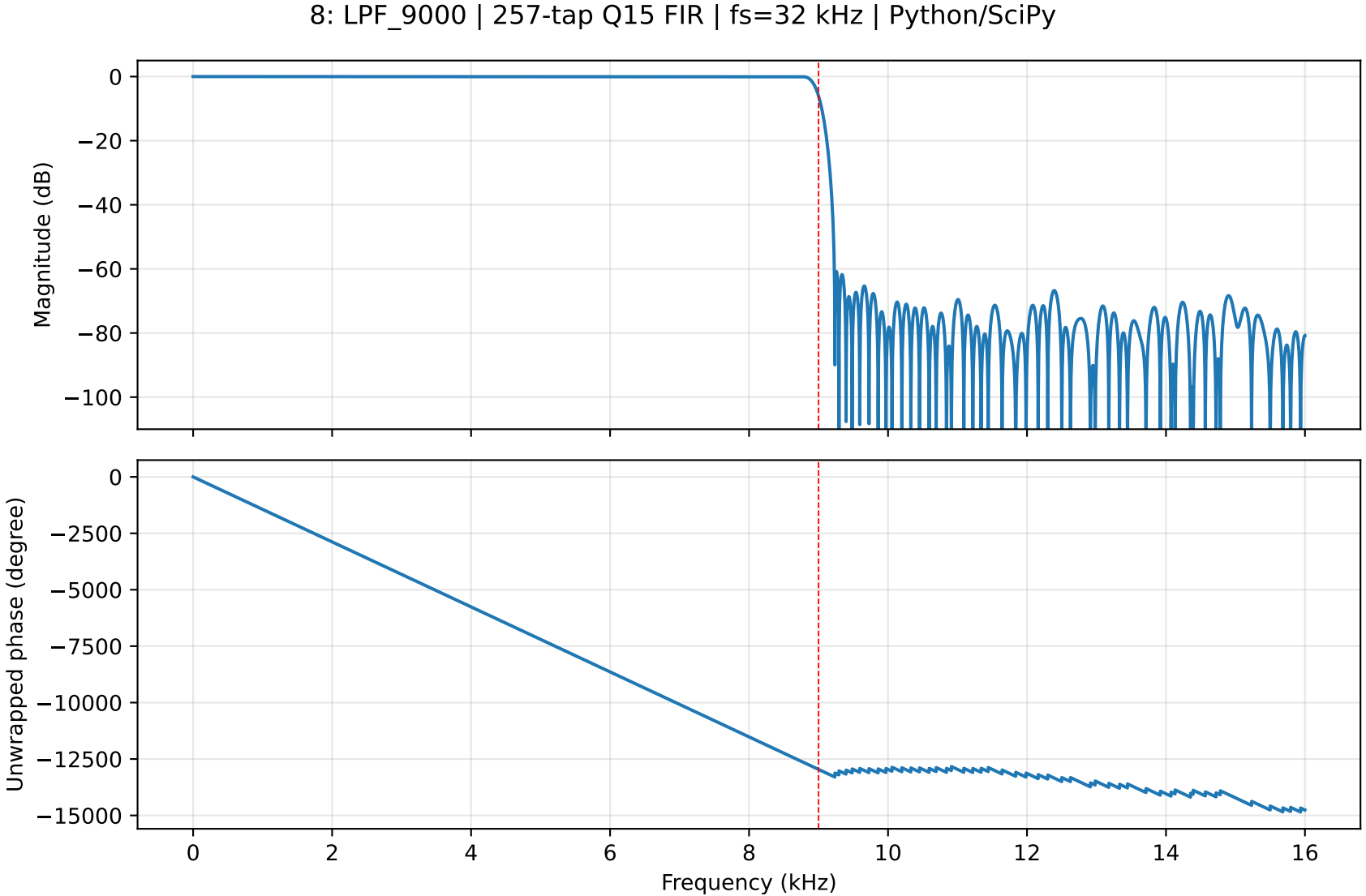

3: LPF_4000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
−12000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

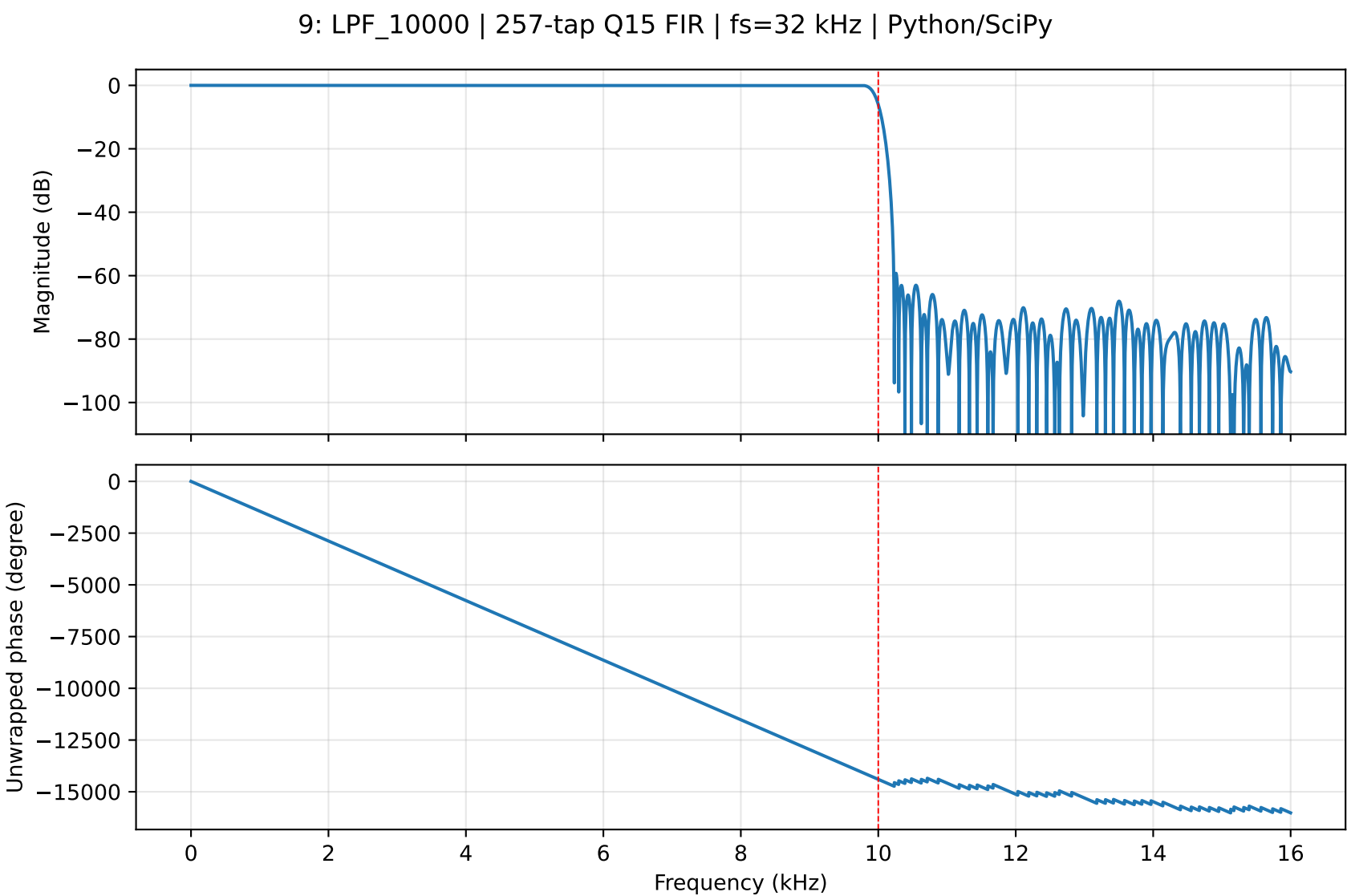

4: LPF_5000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
−12000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

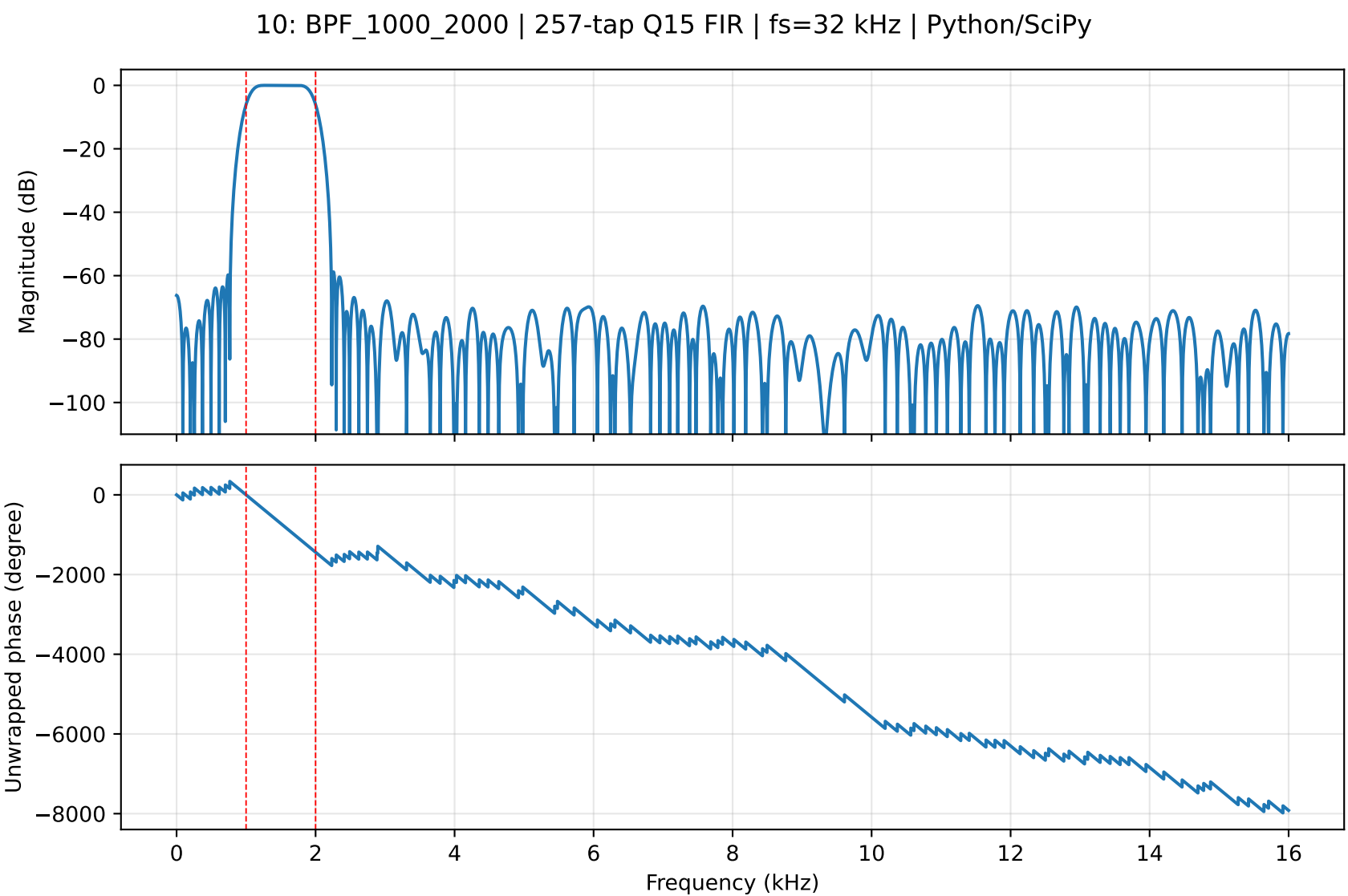

5: LPF_6000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
−12000
−14000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

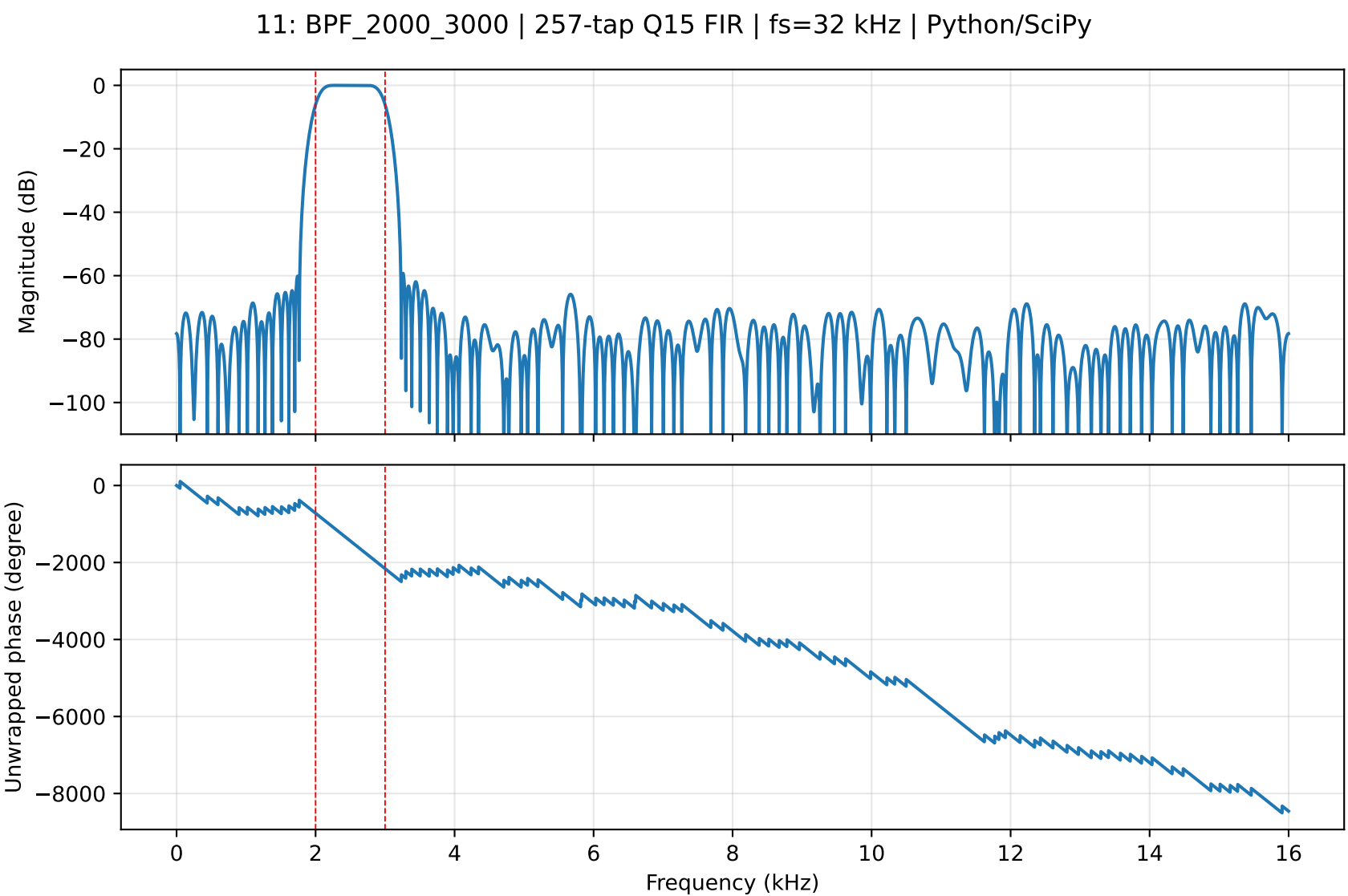

6: LPF_7000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2500
−5000
−7500
−10000
−12500
−15000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

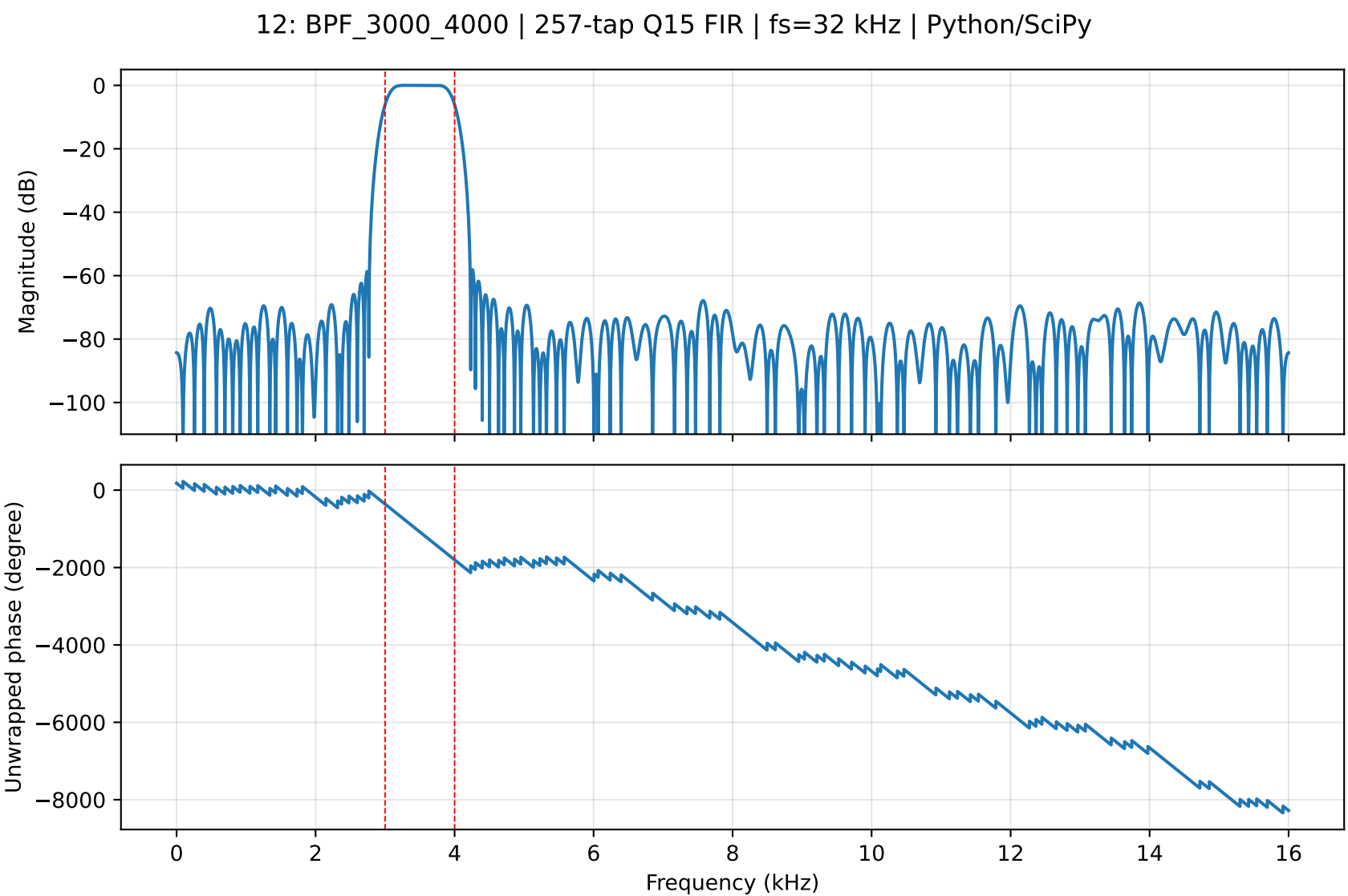

7: LPF_8000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
−12000
−14000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

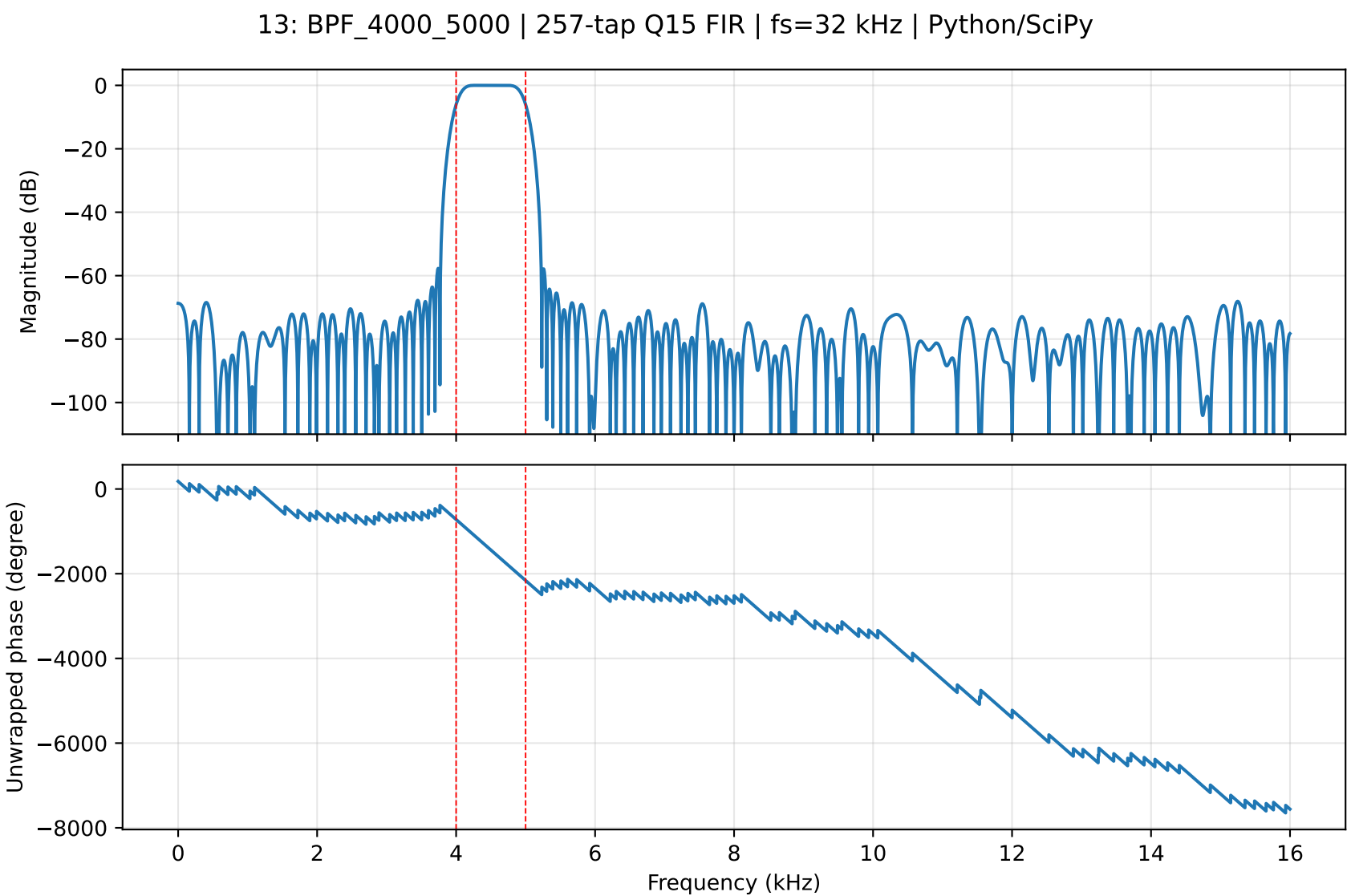

8: LPF_9000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2500
−5000
−7500
−10000
−12500
−15000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

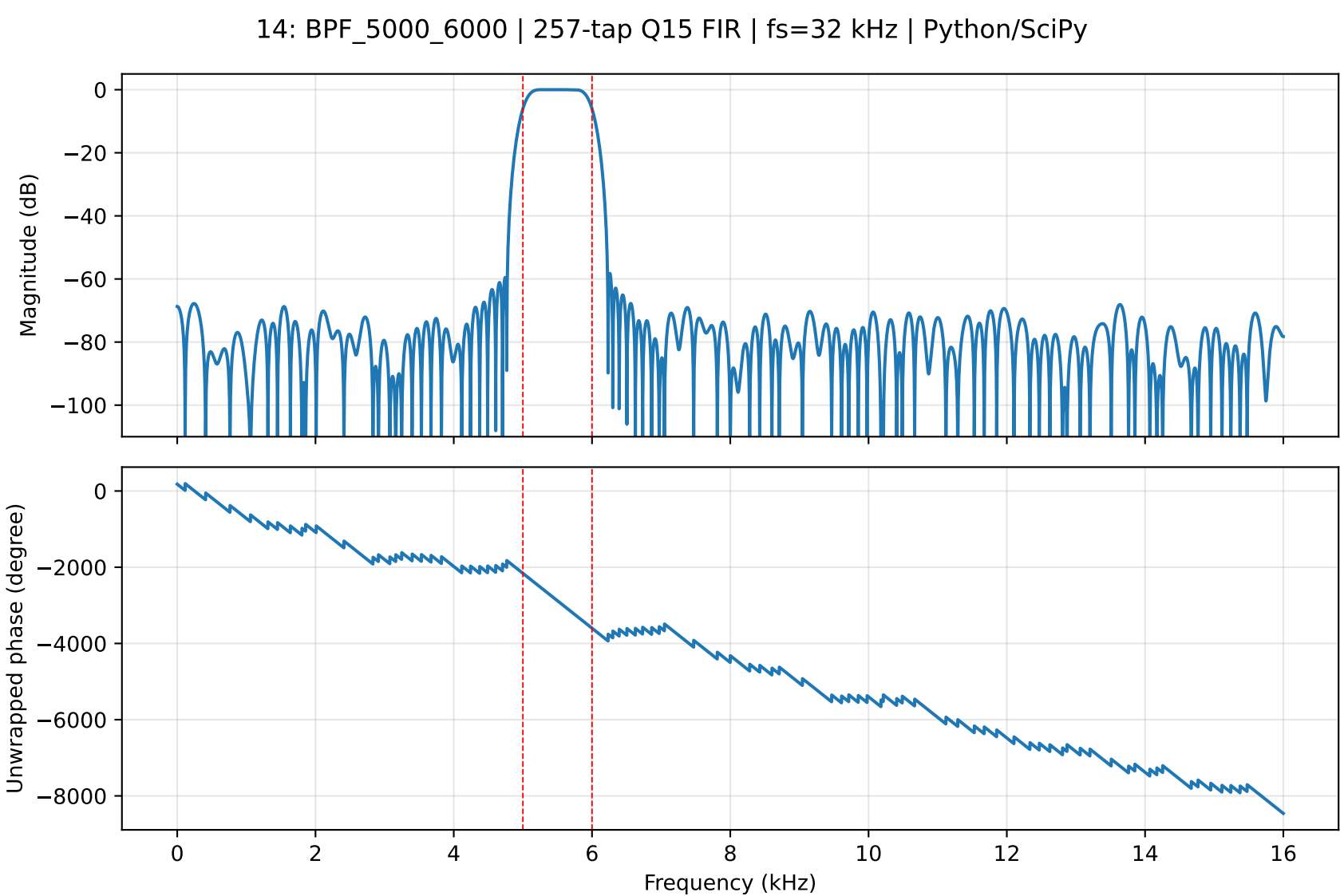

9: LPF_10000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2500
−5000
−7500
−10000
−12500
−15000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

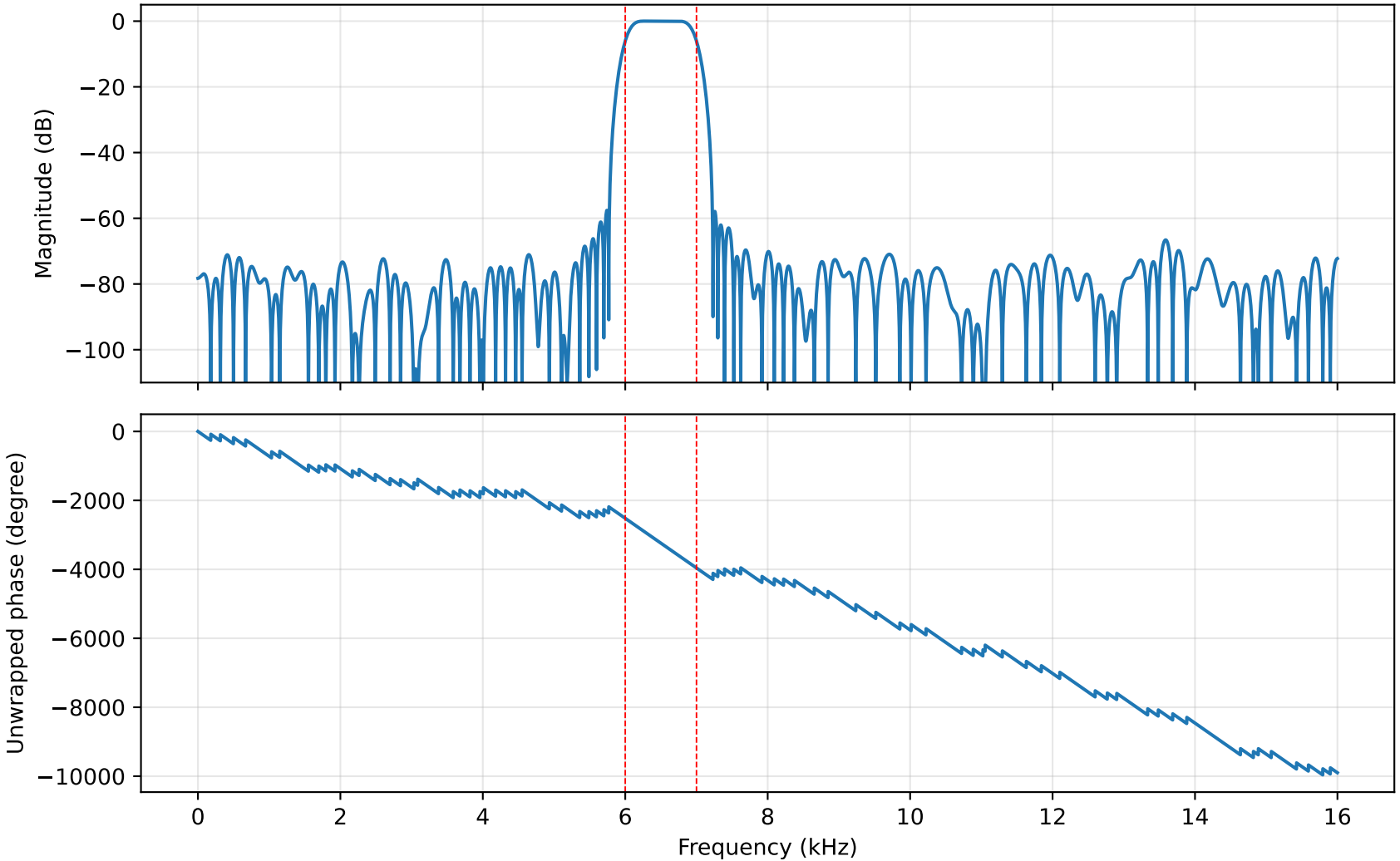

10: BPF_1000_2000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

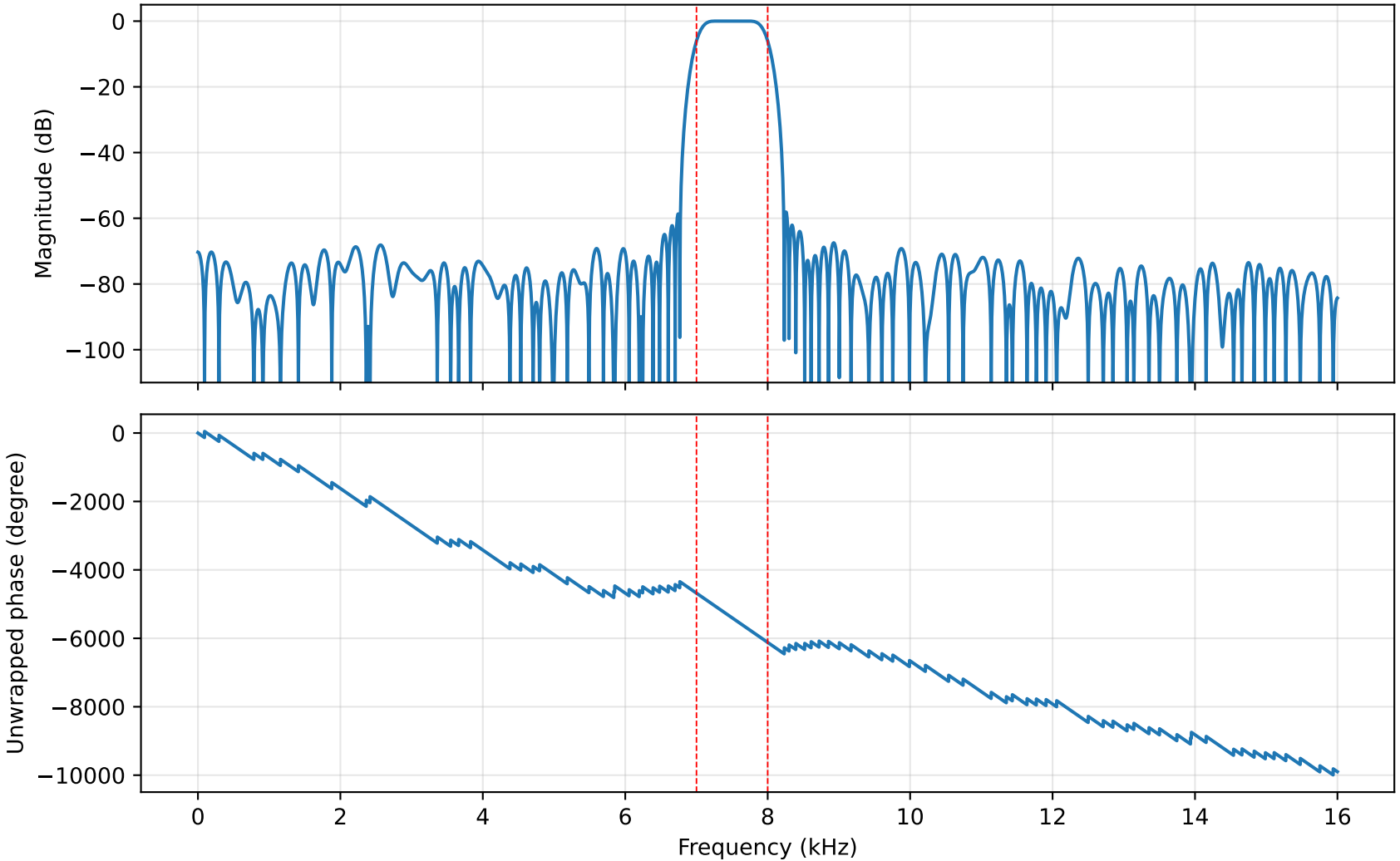

11: BPF_2000_3000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

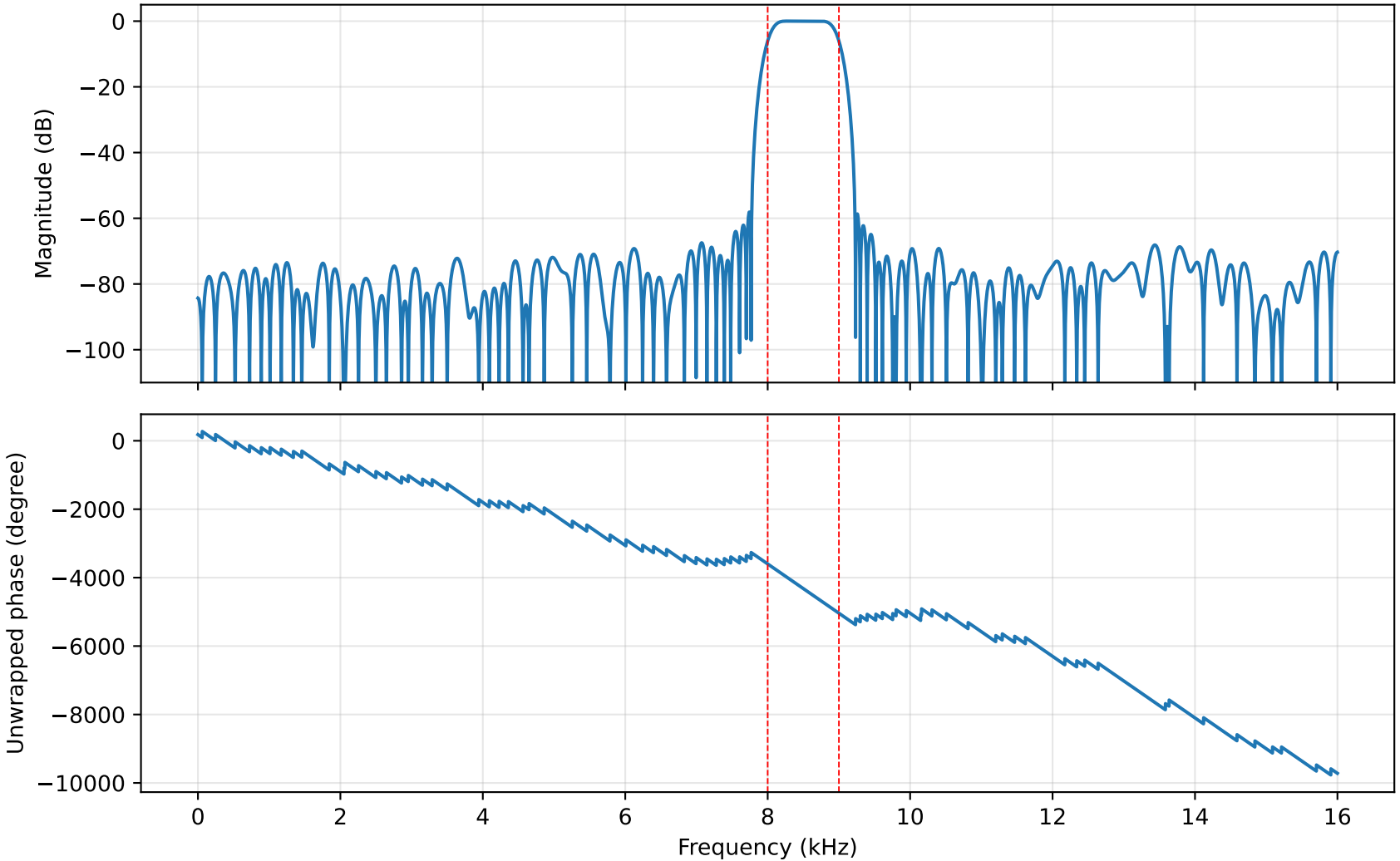

12: BPF_3000_4000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

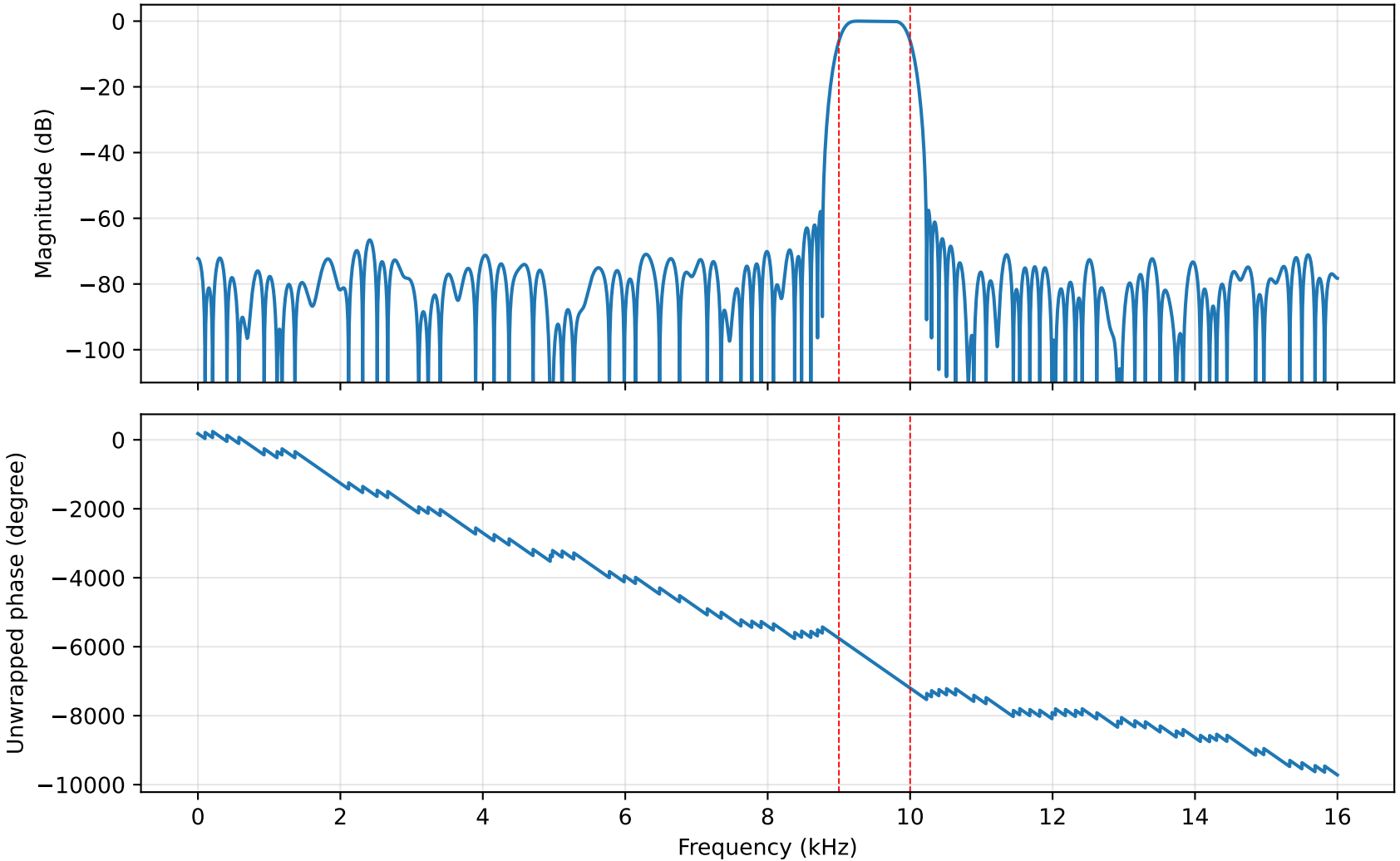

13: BPF_4000_5000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

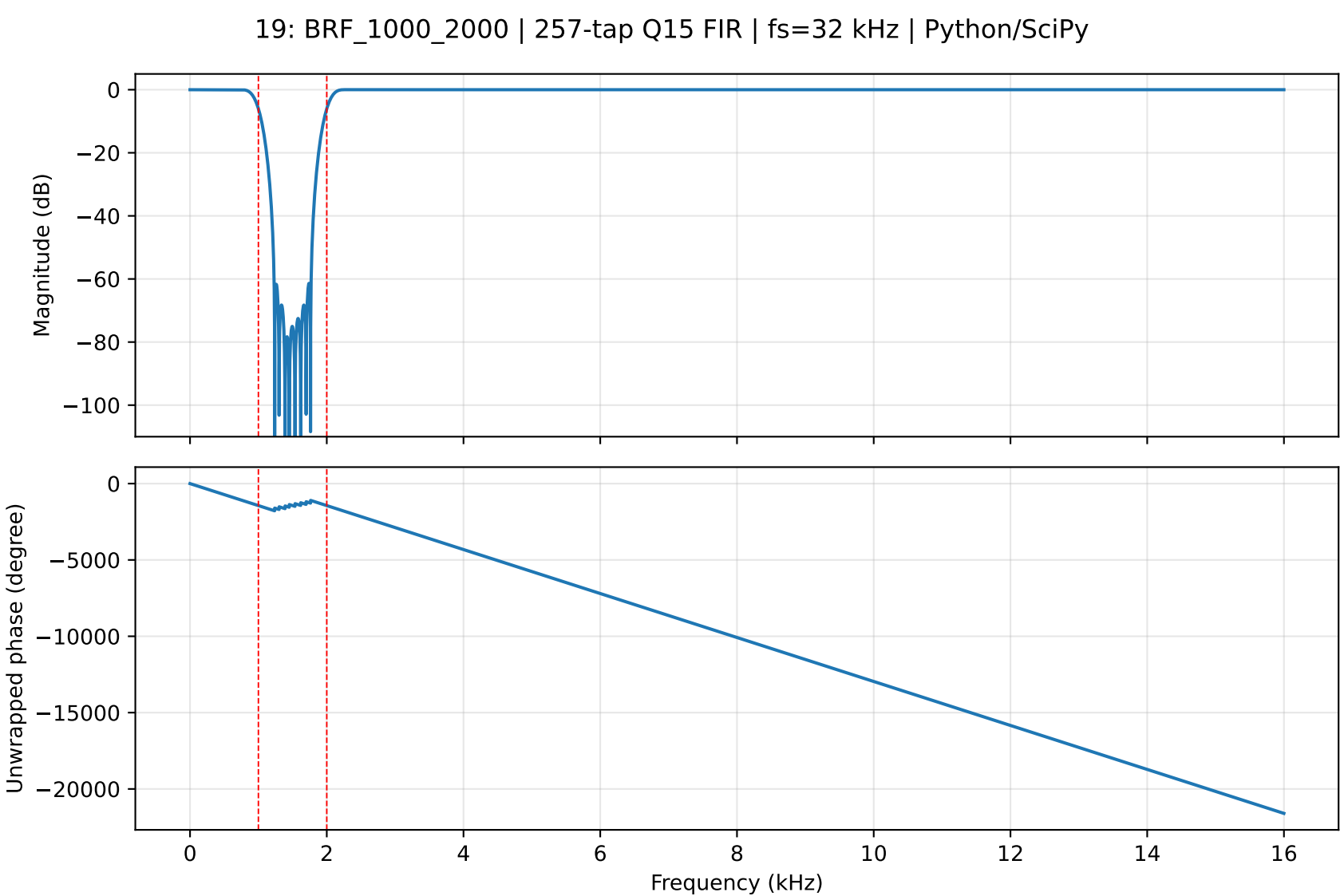

14: BPF_5000_6000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

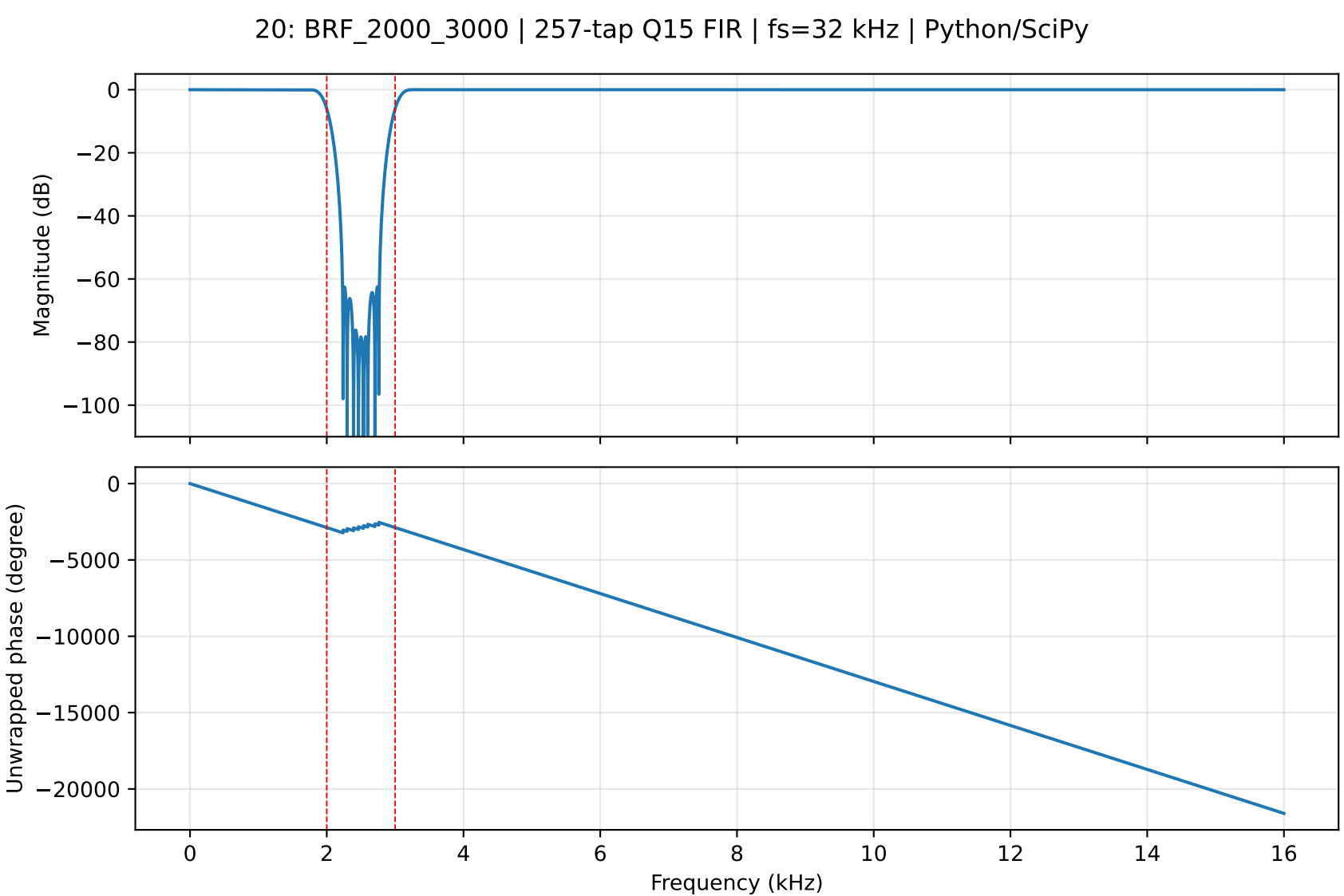

15: BPF_6000_7000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

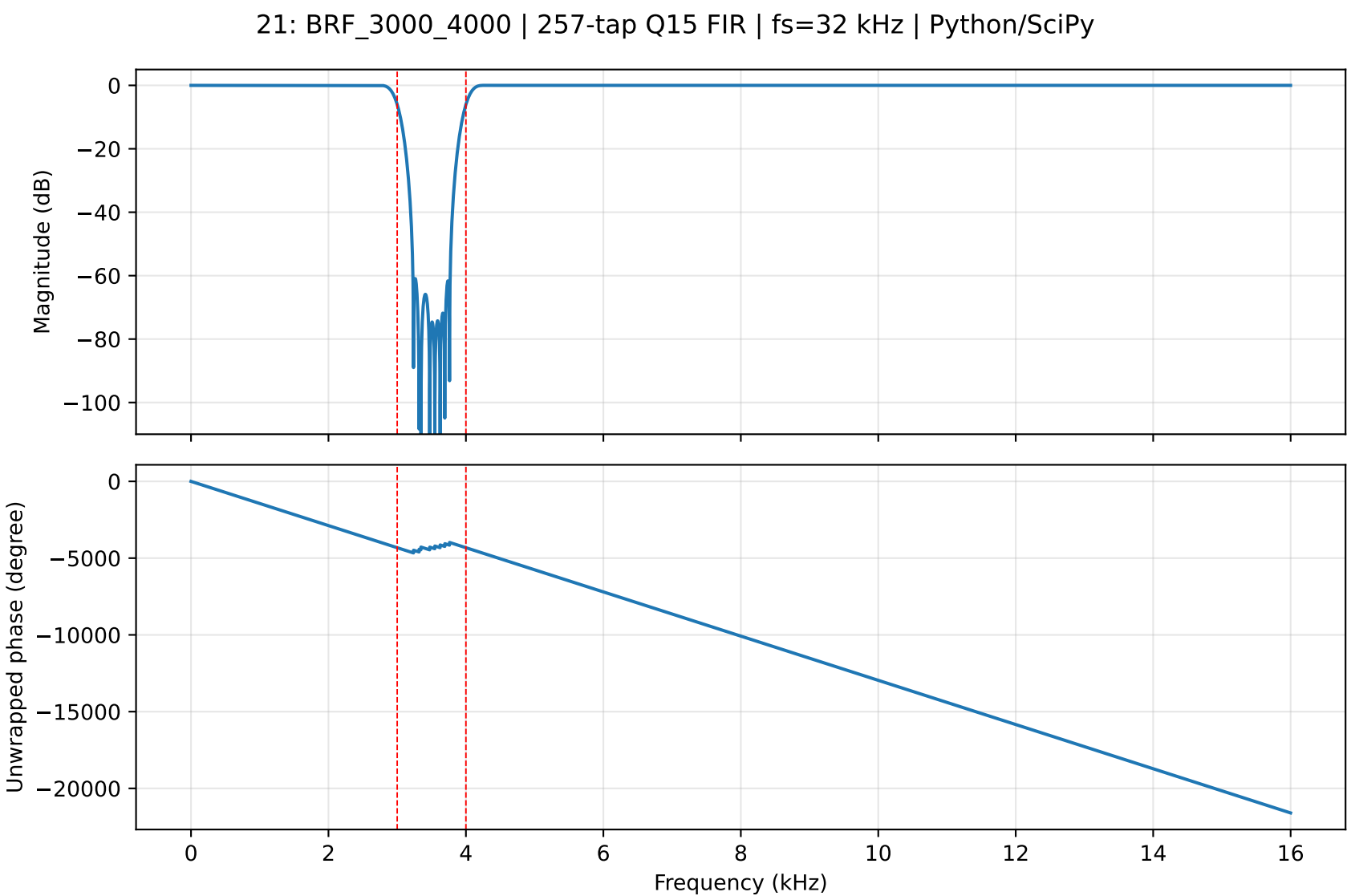

16: BPF_7000_8000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

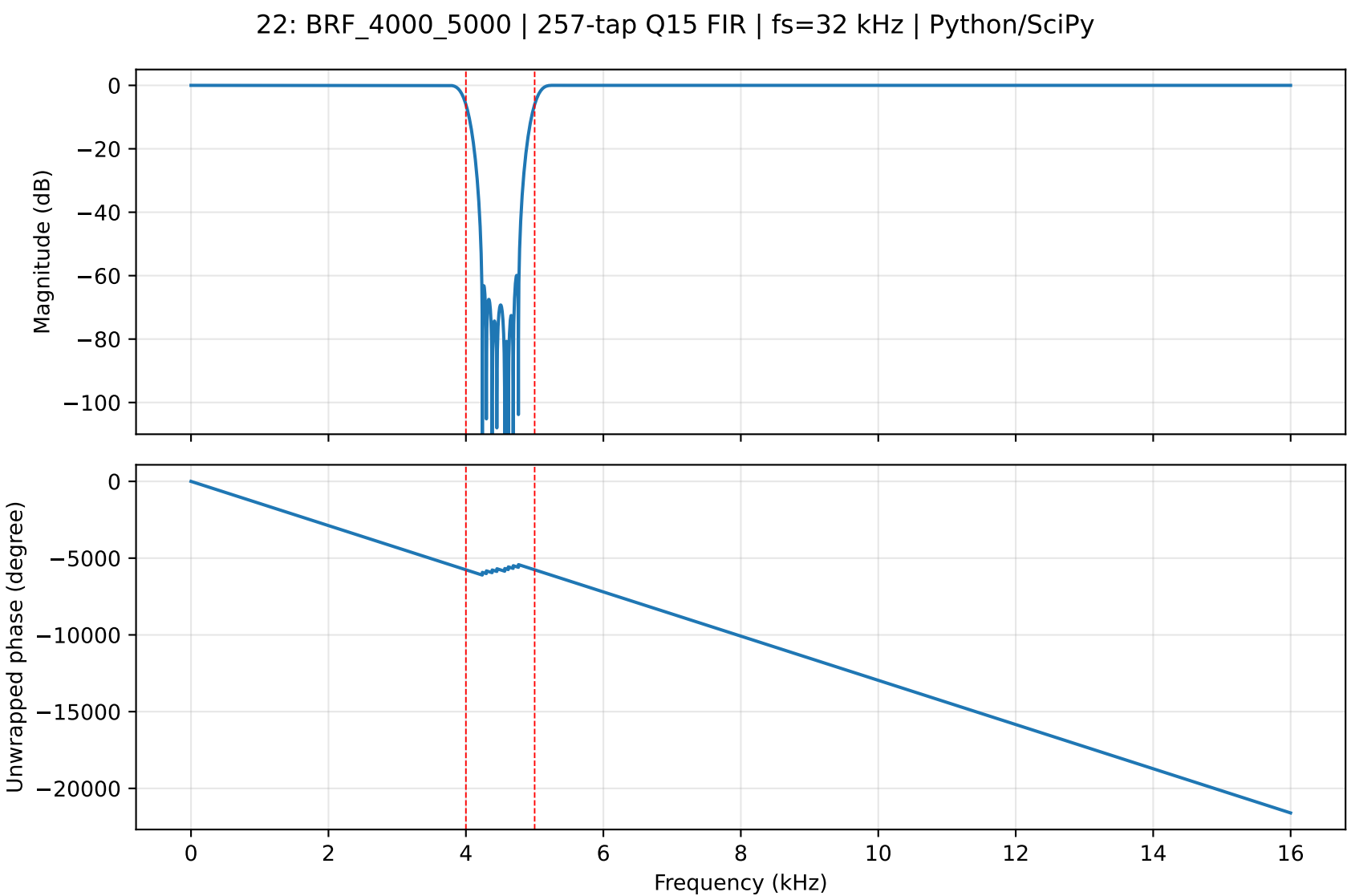

17: BPF_8000_9000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

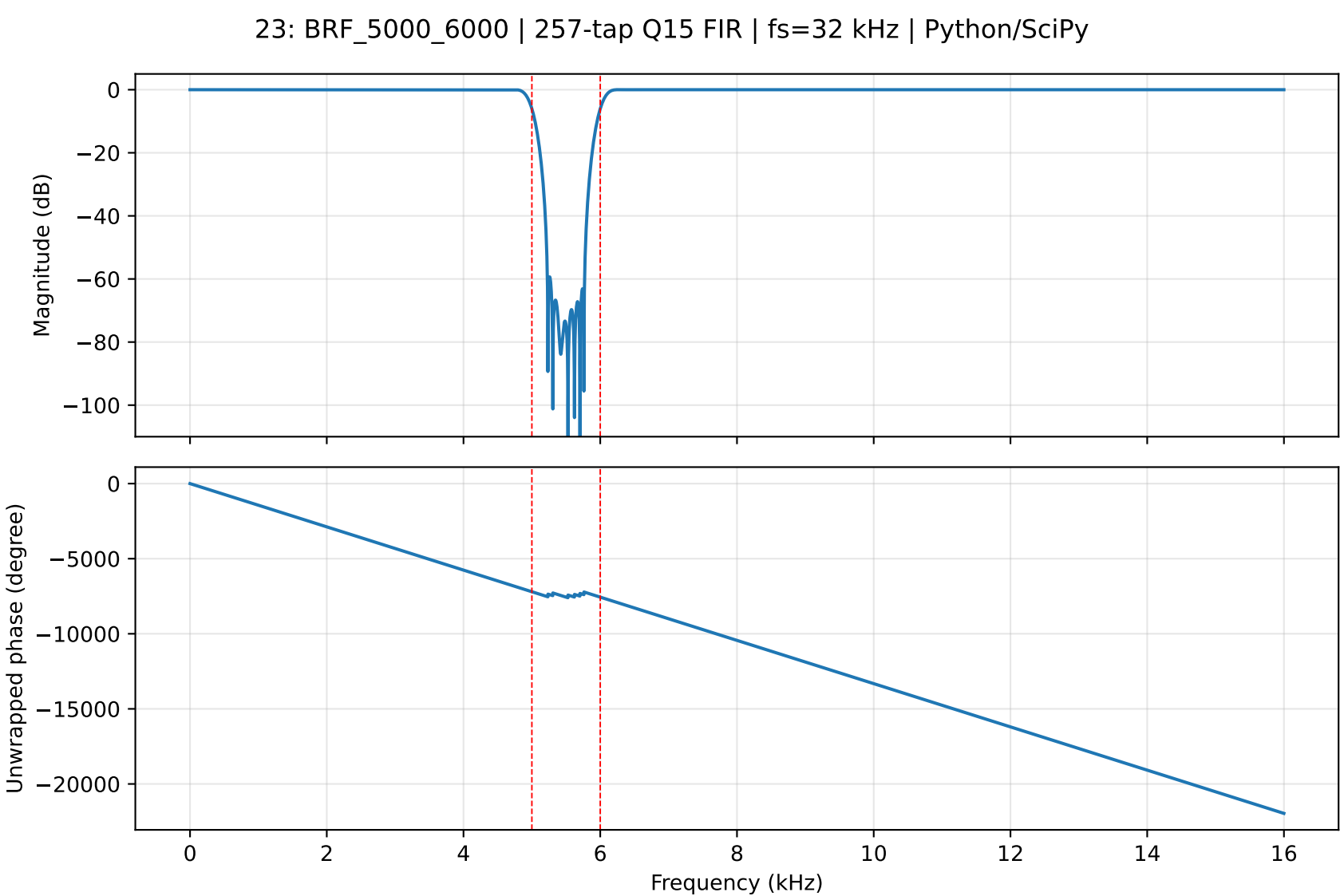

18: BPF_9000_10000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−2000
−4000
−6000
−8000
−10000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

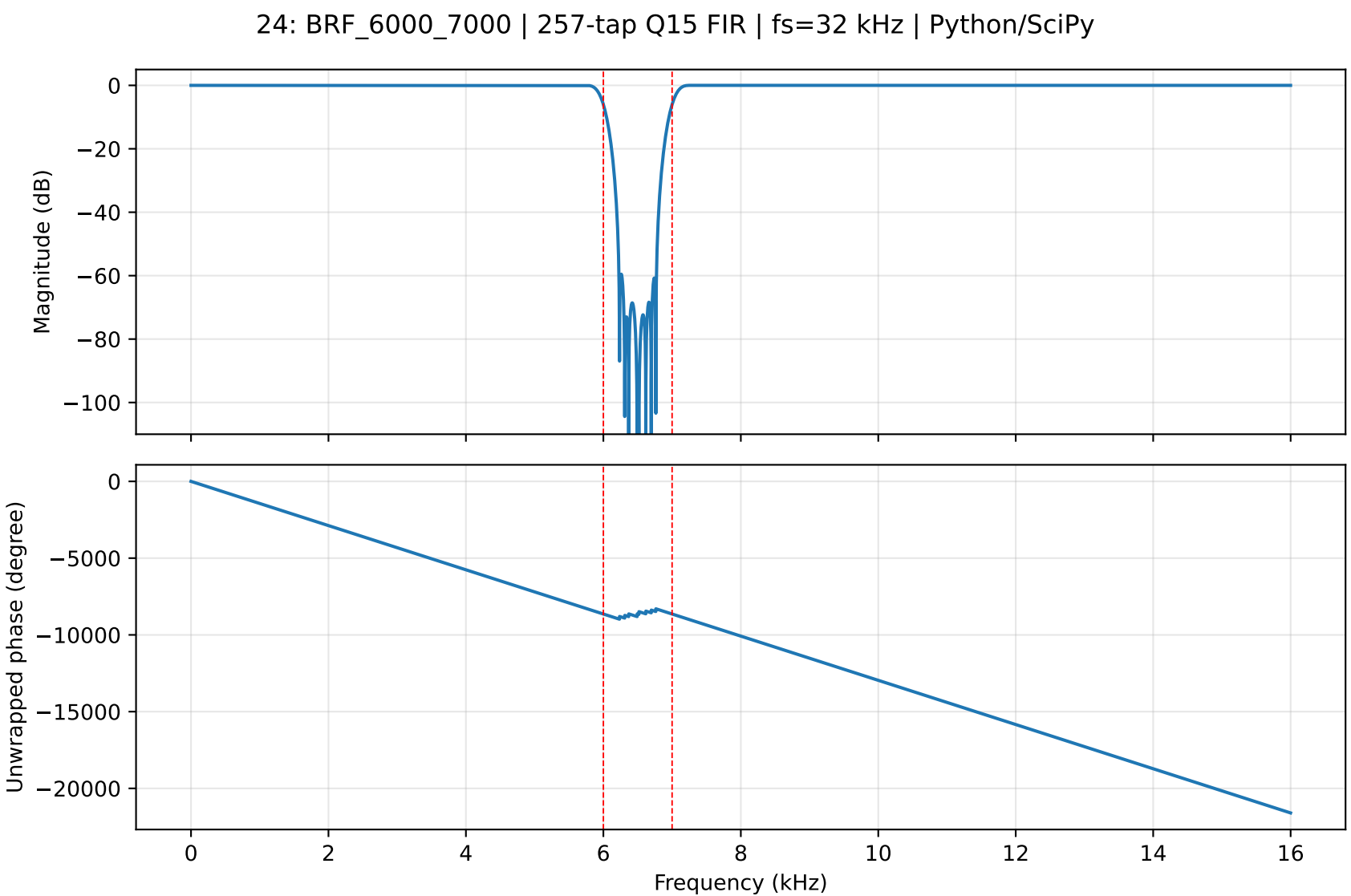

19: BRF_1000_2000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

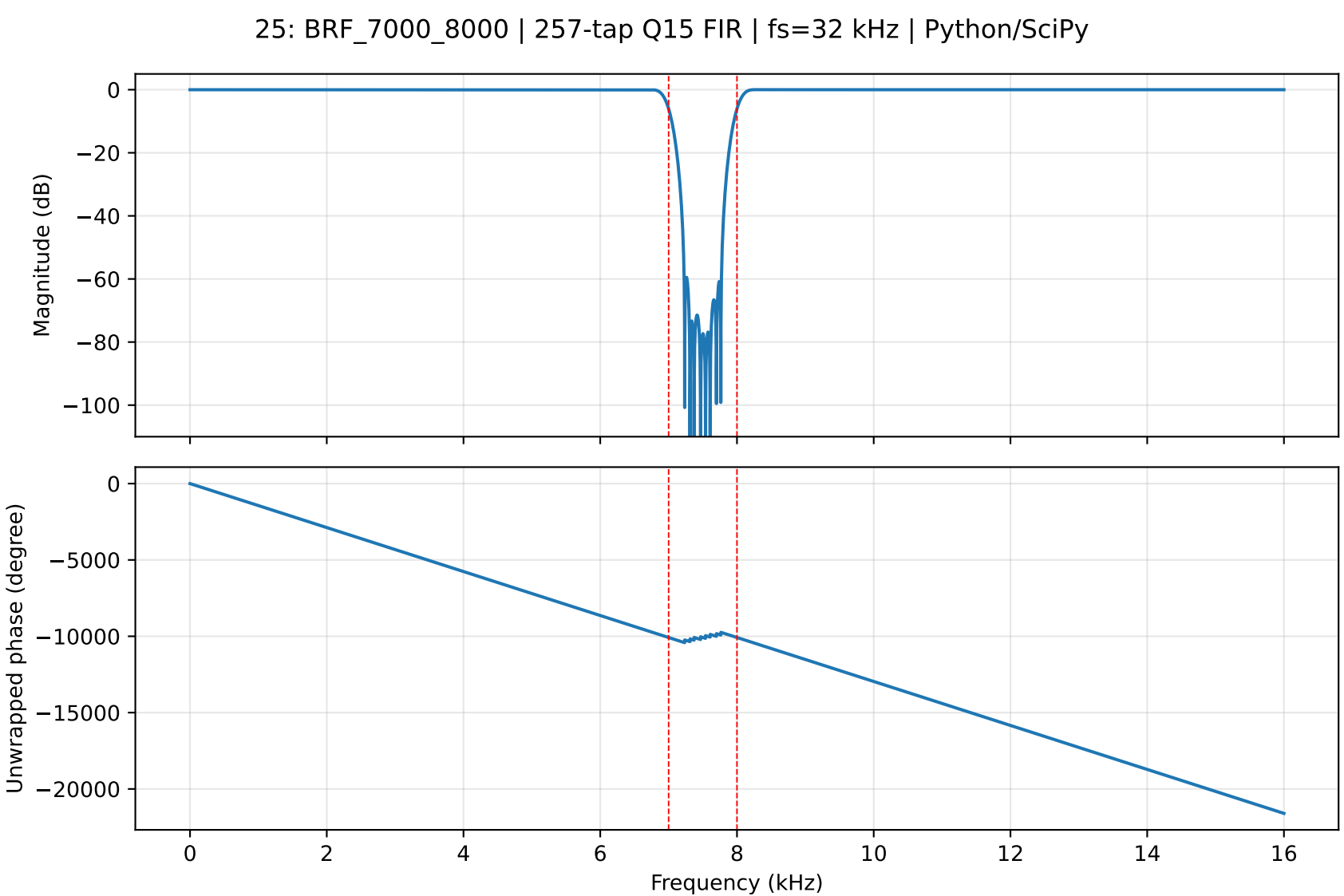

20: BRF_2000_3000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

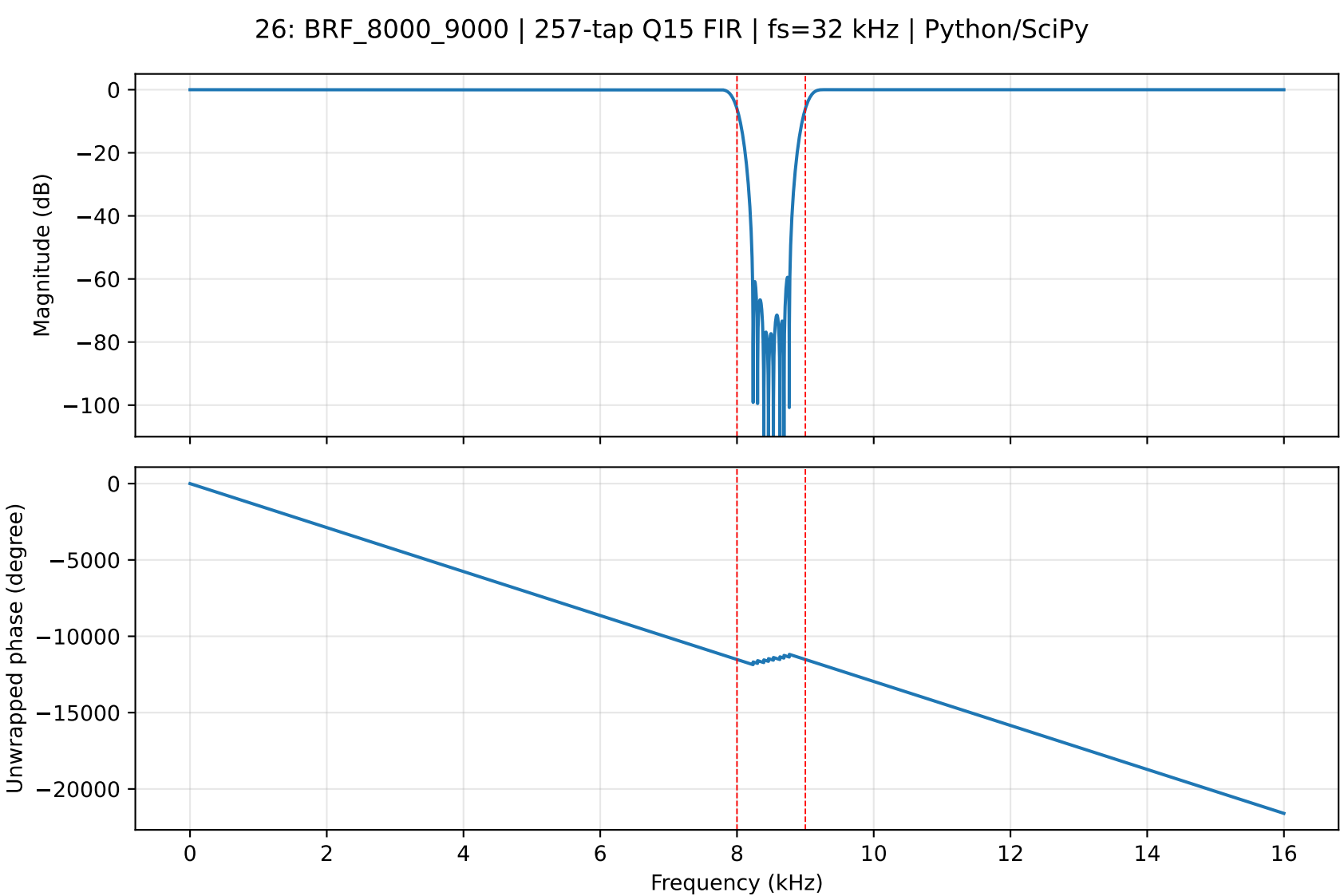

21: BRF_3000_4000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)

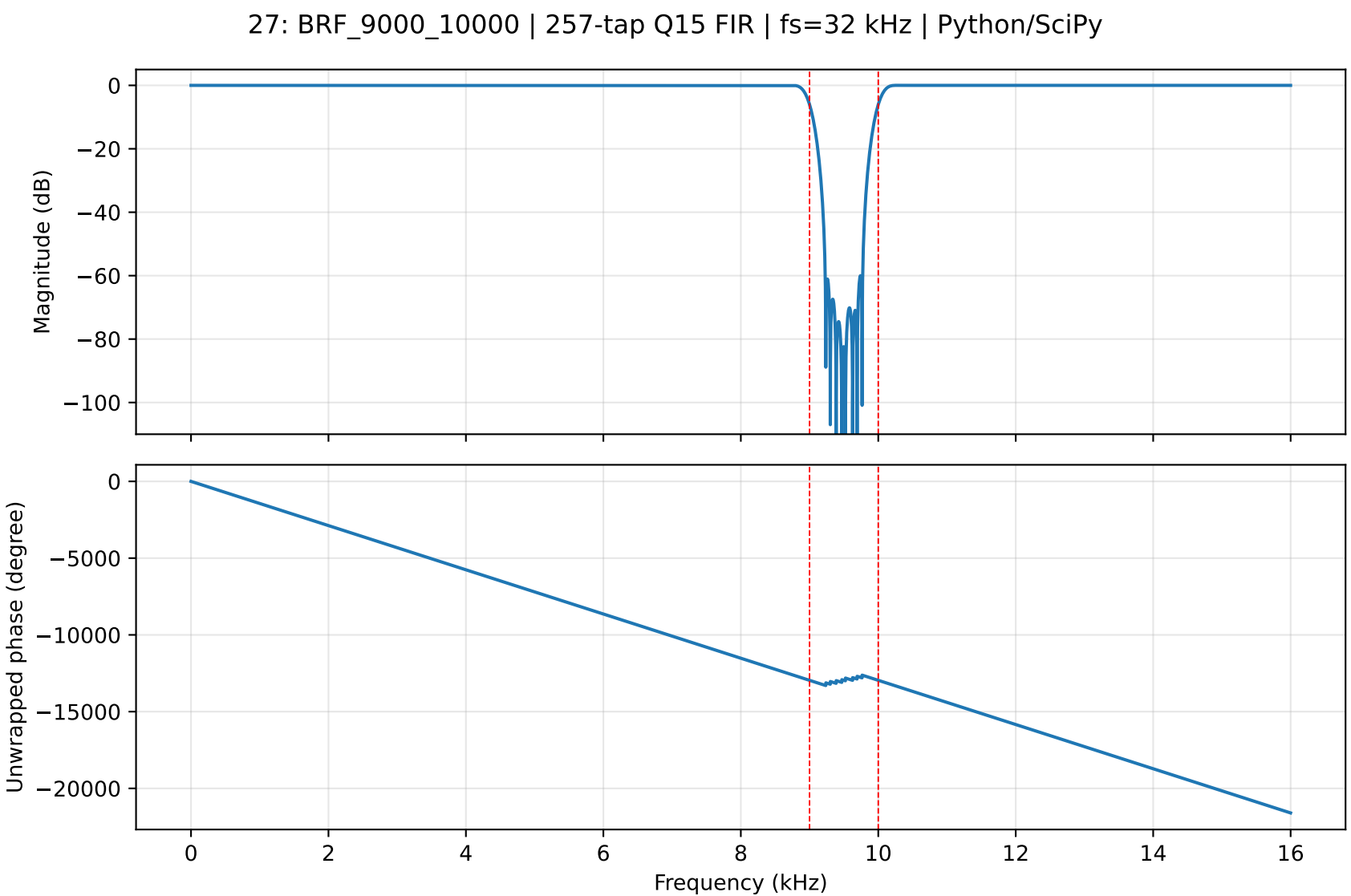

22: BRF_4000_5000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)


23: BRF_5000_6000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)


24: BRF_6000_7000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)


25: BRF_7000_8000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)


26: BRF_8000_9000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)


27: BRF_9000_10000 | 257-tap Q15 FIR | fs=32 kHz | Python/SciPy
Magnitude (dB)
0
−20
−40
−60
−80
−100
Unwrapped phase (degree)
0
−5000
−10000
−15000
−20000
0
2
4
6
8
10
12
14
16
Frequency (kHz)